POST-INCIDENT ANALYSIS REPORT

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1. Objective
The objective of this task is to perform a post-incident analysis for a phishing attack by identifying the root cause, analyzing contributing factors, and calculating SOC performance metrics such as Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR).

2. Tools Used
- Google Sheets – Root Cause Analysis and metrics calculation
- draw.io – Fishbone Diagram creation

3. Incident Description
A phishing email was delivered to a user, leading to unauthorized access after the user clicked a malicious link. The incident highlights gaps in email security controls and user awareness.

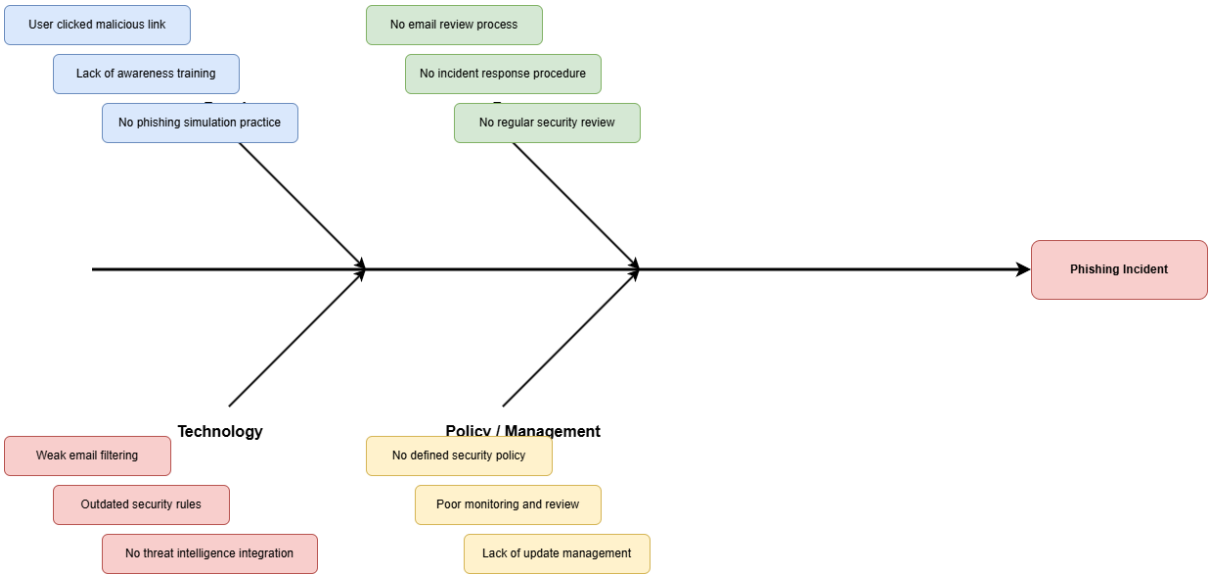
4. Root Cause Analysis (5 Whys)
The 5 Whys technique was used to identify the underlying cause of the incident.

Question	Answer
Why was email opened?	User clicked malicious link
Why was link clicked?	Email appeared legitimate
Why did it appear legitimate?	Weak email filtering allowed phishing email
Why was filtering weak?	No advanced phishing detection rules
Why were rules missing?	Security policies were not updated

Root Cause:
Lack of advanced email filtering mechanisms and outdated security policies enabled the phishing attack.

5. Fishbone Diagram Analysis

A Fishbone (Ishikawa) diagram was created using draw.io to categorize contributing factors.



Key Cause Categories:

- **People**
  - Lack of phishing awareness
  - Insufficient user training
- **Process**
  - No email verification procedures
  - Weak incident response workflow
- **Technology**
  - Ineffective email filtering system
  - Absence of anti-phishing tools
- **Policy**
  - Outdated security policies
  - No regular awareness programs

This structured approach helped identify multiple contributing factors beyond the immediate cause.

6. SOC Metrics Calculation
The following metrics were calculated for the incident:

Metric	Value
Detection Time	2 Hours
Response Time	4 Hours
MTTD	2 Hours
MTTR	4 Hours

Interpretation:
- **MTTD (2 hours)** indicates a delay in identifying the incident.
- **MTTR (4 hours)** reflects moderate response efficiency.

7. Analysis
The analysis revealed that the primary weakness was in preventive controls rather than response capability. While the response was completed within a reasonable timeframe, earlier detection and stronger filtering mechanisms could have prevented the incident entirely.

8. Lessons Learned
- Implement advanced email filtering solutions
- Conduct regular phishing awareness training
- Update and enforce security policies
- Improve monitoring and alerting systems
- Introduce automation for faster response

9. Metrics Summary
The phishing incident had a Mean Time to Detect (MTTD) of 2 hours and a Mean Time to Respond (MTTR) of 4 hours. Detection delays indicate gaps in monitoring, while response time shows moderate efficiency. Enhancing detection mechanisms and automating response processes can significantly improve SOC performance.